EXPERIMENT 6:

Aim

To implement and simulate the classical AI Vacuum World problem using a reflex agent model in Python to clean a multi-room discrete environment efficiently.

Algorithm

1. Initialize Environment: Set the agent's initial location and the dirt status of all rooms, along with a cost counter of 0.
2. Check Goal State: Evaluate if all rooms are clean. If true, terminate the simulation.
3. Perceive and Decide:
    - If the current room is dirty, execute Suck, update the room to clean, and increment cost.
    - If clean, move to the alternate room, update the location, and increment cost.
4. Iterate: Repeat steps 2 and 3 until the environment is fully cleaned.

CODE:

```
class VacuumWorld:
    def __init__(self, location, status_a, status_b):
        self.location = location
        self.rooms = {'A': status_a, 'B': status_b}
        self.cost = 0

    def step(self):
        if self.rooms[self.location]:
            print(f"Room {self.location} is DIRTY. Action: Suck")
            self.rooms[self.location] = False
            self.cost += 1
        else:
            next_location = 'B' if self.location == 'A' else 'A'
            print(f"Room {self.location} is clean. Action: Move to {next_locatio
            self.location = next_location
            self.cost += 1

    def is_solved(self):
        return not self.rooms['A'] and not self.rooms['B']

world = VacuumWorld(location='A', status_a=True, status_b=True)

step_count = 0
while not world.is_solved() and step_count < 10:
    print(f"--- Step {step_count + 1} (Location: {world.location}, Rooms: {world
    world.step()
    step_count += 1

print(f"\nEnvironment cleaned in {world.cost} steps!")
```

```
============== RESTART: C:/Users/Admin/OneDrive/Documents/aiexp6.py
--- Step 1 (Location: A, Rooms: {'A': True, 'B': True}) ---
Room A is DIRTY. Action: Suck
--- Step 2 (Location: A, Rooms: {'A': False, 'B': True}) ---
Room A is clean. Action: Move to B
--- Step 3 (Location: B, Rooms: {'A': False, 'B': True}) ---
Room B is DIRTY. Action: Suck

Environment cleaned in 3 steps!
```

Result

The simulation successfully initialized the vacuum agent in a two-room environment with dirt present in both locations. The agent executed sequential perceptual checks, performed Suck actions to clean dirty rooms, and navigated between rooms until all locations reached a clean state.

Conclusion

The program successfully demonstrated the decision-making process of a reflex-based agent in the Vacuum World. By evaluating current percepts at each time step, the agent determined appropriate actions (Suck or Move) to achieve the goal state, highlighting foundational artificial intelligence concepts including state representation and action execution.